Modeling and Stress-Testing Housing Affordability in Canada

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***Abstract*—Housing affordability has become a major economic issue in Canada [1, 2]. However, it is still unclear how much interest rates, income levels, and housing supply each contribute to the problem. This study builds a data-driven model to measure and predict housing affordability across Canadian provinces and major metropolitan areas, including Toronto, Vancouver, and Montreal.**

Affordability is measured using two indicators: the Price-to-Income Ratio (PIR) and the Mortgage Burden Ratio (MBR). Housing price data is obtained from the Canadian Real Estate Association through its MLS® Home Price Index. Income and demographic data are sourced from Statistics Canada [3]. Interest rate data is collected from the Bank of Canada [4]. Housing supply data, including housing starts, is provided by Canada Mortgage and Housing Corporation [5].

The study applies panel regression analysis to estimate how affordability changes in response to interest rates, income growth, and housing supply. Panel econometric methods are widely used to analyze economic relationships across regions and time [6]. A policy shock analysis is also conducted to evaluate the impact of recent interest rate increases.

In addition, scenario simulations are performed to test how affordability would change under a 1% interest rate increase, a 1% rate cut, 3% income growth, and a 10% increase in housing supply. These simulations help compare the relative effectiveness of different economic and policy levers.

The results provide both academic insight and practical value. The findings support risk assessment for financial institutions, capital planning for developers, and evidence-based policy design. By combining quantitative modeling with applied scenario analysis, this study offers a structured framework for understanding and stress-testing housing affordability in Canada.

I. Introduction

Housing affordability in Canada has become a major economic and social concern [1, 2]. Home prices have increased rapidly in many regions, while borrowing costs have changed significantly in recent years. Research in housing economics shows that housing prices are influenced by a combination of interest rates, income growth, and housing supply constraints [7].

However, the relative impact of different economic forces on affordability remains unclear.

Several key questions motivate this study:

- How strongly do rising interest rates affect housing affordability?
- Does income growth cause higher housing prices?
- Does increasing housing supply reduce affordability pressure?
- Are major metropolitan areas more sensitive to interest rate changes than other regions?

Answering these questions is important for both academic and real-world decision-making. Financial institutions need to assess mortgage risk exposure. Developers must evaluate supply responsiveness. Policymakers require evidence on which economic lever has the greatest impact.

This paper addresses these questions using panel econometric modeling and scenario simulation. Panel data methods are commonly used in empirical economic research to analyze relationships that vary across regions and time [6].

Affordability is measured using the Price-to-Income Ratio (PIR) and the Mortgage Burden Ratio (MBR), and the sensitivity of these measures to interest rates, income, and supply is formally estimated.

II. Institutional Background and Data Foundation

This study models housing affordability as a system made up of four interacting components:

1) Housing Prices
2) Interest Rates
3) Household Income
4) Housing Supply

A. *Canadian Real Estate Association (CREA): Measuring Price Dynamics*

The Canadian Real Estate Association (CREA) publishes the MLS® Home Price Index (HPI), which tracks changes in housing prices over time. The HPI is widely used to measure housing price dynamics in Canadian real estate markets because it adjusts for differences in housing type and quality [2].

Unlike simple average sale prices, the HPI controls for compositional changes in the mix of homes sold. This makes it more suitable for time-series modeling because it reduces noise caused by changes in the types of properties transacted in the market.

In this study, housing price data is used to construct the price-to-income ratio:

$$PIR_{it} = \frac{Price_{it}}{Income_{it}}$$

Where:

- PIR_{it} is the price-to-income ratio in region i at time t,
- Price_{it} is the median home price in region i at time t,
- Income_{it} is the median household income in region i at time t.

The price-to-income ratio is a commonly used indicator of housing affordability in economic research [7]. Higher values of the ratio indicate that housing prices are large relative to household income.

In the empirical framework, housing prices act as a key feature in predicting affordability. A higher PIR means housing is less affordable.

For example, if the PIR equals 8, this means the median home price is eight times the median household income. Higher values indicate greater affordability pressure.

B. Bank of Canada: Interest Rate Transmission

The Bank of Canada sets the policy interest rate, which influences borrowing costs across the economy.

Mortgage rates are strongly affected by changes in the policy rate. When interest rates increase, mortgage payments rise even if housing prices remain constant.

In this study, interest rates enter the model through the Mortgage Burden Ratio:

$$MBR_{it} = \frac{12M_{it}}{Income_{it}}$$

Where M_{it} is the monthly mortgage payment, calculated as:

$$M_{it} = P_{it} \cdot \frac{r_t(1+r_t)^n}{(1+r_t)^n - 1}$$

Where:

- P_{it} is the loan principal (home price minus down payment),
- r_t is the monthly mortgage interest rate,
- n is the total number of monthly payments.

This formula translates macroeconomic policy into household-level financial stress. in modeling terms, interest rates act as a shock variable. When interest rates increase, monthly payments increase, even if home prices remain constant. This directly increases the Mortgage Burden Ratio. From a machine learning viewpoint, the interest rate is an exogenous input that shifts the affordability distribution.

C. Statistics Canada: Income and Demand Capacity

Statistics Canada provides data on household income and population [3].

Income is crucial because housing affordability depends not only on housing prices but also on household purchasing power.

Income enters directly into both affordability measures:

- PIR (denominator)
- MBR (denominator)

Population data is also used to approximate demand pressure. Rapid population growth may increase housing demand and affect price trends [1].

In modeling terms, income and population represent the **demand-side features** of the housing system.

In predictive modeling language, these variables help explain variation in affordability across regions and over time.

D. Canadian Mortgage and Housing Corporation (CMHC): Supply Constraints

The Canada Mortgage and Housing Corporation (CMHC) provides data on housing starts and construction activity [5].

Housing starts are commonly used as a proxy for housing supply in empirical housing market research.

Supply plays a structural role in affordability. If supply increases faster than demand, price pressure may decrease. If supply is limited, price growth may accelerate [7].

In the econometric model, supply enters as:

$$\ln(A_{it}) = \alpha_i + \gamma_t + \beta_1 Rate_t + \beta_2 \ln(Income_{it}) + \beta_3 \ln(Supply_{it}) + \varepsilon_{it} \tag{1}$$

Where:

- A_{it} represents the affordability measure (either PIR or MBR) in region i at time t,
- α_i represents region fixed effects, which control for permanent differences across provinces or cities,
- γ_t represents time fixed effects, which control for national shocks affecting all regions,
- $Rate_t$ is the interest rate at time t,
- $Income_{it}$ is median household income,
- $Supply_{it}$ represents housing supply (e.g., housing starts),
- ε_{it} is the error term.

This model estimates how affordability changes in response to interest rates, income, and housing supply, while controlling for regional differences and national economic conditions.

The logarithmic form allows the coefficients to be interpreted as elasticities, which is common in empirical economic modeling [6].

This means that β_2 and β_3 measure the percentage change in affordability resulting from a one percent change in income or supply.

Using the natural logarithm helps stabilize variance and allows interpretation in percentage terms, which improves economic interpretation.

E. Example Affordability Calculations Used in the Paper

Price-to-Income Ratio

PIR is one of the most commonly used indicators of housing affordability in economic research [2].

- Typical home price in the region: $700,000
- Median household income: $120,000 per year

Then:

$$PIR = \frac{700{,}000}{120{,}000} = 5.83$$

In simple terms, homes cost about 5.8 times yearly income.

Mortgage Burden Ratio with Amortization

MBR measures how heavy the mortgage payment is compared to income. We compute the monthly mortgage payment using the standard amortization formula:

$$M = P \cdot \frac{r(1+r)^n}{(1+r)^n - 1}$$

- Home price: \$700,000
- Down payment: 20% $\rightarrow$ \$140,000
- Mortgage principal: $P = 700{,}000 - 140{,}000 = 560{,}000$
- Annual interest rate: 5%
- Monthly interest rate: $r = 0.05/12 = 0.004167$
- Amortization: 25 years $\rightarrow n = 25 \times 12 = 300$ monthly payments

Using the formula:

$$M \approx 3{,}274$$

Annual mortgage payments:

$$12M \approx 39{,}284$$

If household income is \$120,000 per year:

$$MBR = \frac{39{,}284}{120{,}000} = 0.327 = 32.7\%$$

This household spends about 33% of its income on mortgage payments.

III. Econometric Model

This section explains the statistical model used to measure housing affordability in Canada. The goal of the model is to estimate how affordability changes when interest rates or income change.

We use a panel data regression model. Panel data means the dataset contains observations across both time and regions. In this study, the regions are Canadian provinces and the time period covers the years from 2009 to 2023.

Panel fixed-effects models are widely used in applied econometric research to analyze data that vary across both regions and time [6].

A. Model Specification

The main econometric model is written as:

$$\ln(A_{it}) = \alpha_i + \gamma_t + \beta_1 Rate_t + \beta_2 \ln(Income_{it}) + \varepsilon_{it}$$

Where:

- A_{it} represents the affordability measure in region i at time t. In this study, affordability is measured using the Price-to-Income Ratio (PIR).
- $\ln(A_{it})$ is the natural logarithm of the affordability measure.
- $Rate_t$ represents the Bank of Canada policy interest rate.
- $Income_{it}$ represents median household income in region i at time t.
- α_i represents region fixed effects.
- γ_t represents year fixed effects.
- ε_{it} is the error term.

This fixed-effects specification controls for unobserved differences across regions and time that may influence housing affordability [6].

B. Why We Use Logarithms

The dependent variable is expressed in natural logarithm form.

Using logarithms has several advantages:

- It allows the coefficients to be interpreted as percentage changes.
- It reduces the impact of extreme values.
- It makes the relationship between variables more stable.

Log-linear specifications are commonly used in empirical economic research because they allow coefficients to be interpreted as elasticities [6].

For example, if the coefficient on income equals -1.0, this means that a 1% increase in income reduces affordability pressure by approximately 1%.

C. Fixed Effects

The model includes both region and year fixed effects.

Region fixed effects control for permanent differences between provinces. For example, housing markets in British Columbia may behave differently from those in Quebec due to geography, zoning rules, zoning regulations, or long-term demand patterns.

Year fixed effects control for national shocks that affect all regions at the same time. Examples include economic recessions, inflation shocks, or nationwide policy changes.

Including both region and year fixed effects allows the model to isolate how changes within each region over time affect housing affordability while controlling for unobserved differences across regions and national trends [6].

D. Interpretation of Coefficients

The coefficients in the model measure how housing affordability responds to economic changes.

- The coefficient on the interest rate (β_1) measures how affordability changes when borrowing costs increase.
- The coefficient on income (β_2) measures how affordability changes when household purchasing power increases.

If the interest rate coefficient is positive, this suggests that higher interest rates worsen housing affordability.

If the income coefficient is negative, this suggests that higher income improves housing affordability.

These expected relationships follow the economic reasoning described in the theoretical framework.

IV. Data Description

This section describes the data used to estimate the econometric model. The dataset combines information from several Canadian institutions and covers the period from 2009 to 2023.

The objective of the dataset is to measure housing affordability across Canadian regions while tracking changes in interest rates and household income over time.

A. Data Sources

Three main data sources are used in this study: housing price data, household income data, and interest rate data.

Housing Prices

Housing price data are obtained from Statistics Canada through the New Housing Price Index (NHPI) [8].

In this study, the national housing price index is used as a proxy for housing market price dynamics. Although the index is reported at the national level, combining it with province-level income data allows the analysis to capture regional differences in affordability.

Housing price data are originally reported at a monthly frequency. Monthly values are converted into annual averages in order to match the frequency of the income data.

Household Income

Household income data are obtained from Statistics Canada Table 11-10-0190-01 [3].

Income is an important variable because housing affordability depends not only on housing prices but also on household purchasing power.

Interest Rates

Interest rate data are obtained from the Bank of Canada [4]. The policy interest rate is used as the primary measure of borrowing costs in the economy.

Changes in the policy rate influence mortgage interest rates and therefore affect housing affordability.

B. Dataset Construction

The dataset is constructed by merging the three data sources using the year as the common time variable.

The main steps used to build the dataset are:

- Monthly housing price index values are converted into annual averages.
- Income data from Statistics Canada are matched by province and year [3].
- Interest rate data from the Bank of Canada are matched by year [4].
- All datasets are merged to form a panel dataset containing both regional variation and time variation.

The final dataset includes observations for three provinces:

- British Columbia
- Ontario
- Quebec

These provinces are selected because they represent major housing markets in Canada and provide meaningful variation in housing affordability conditions.

C. Key Variables

The main variables used in the empirical analysis are:

- **PIR**: Price-to-Income Ratio. This measures housing affordability by comparing housing prices to household income and is widely used as an affordability indicator in housing research [2].
- **lnPIR**: Natural logarithm of the Price-to-Income Ratio.
- **PolicyRate**: Bank of Canada policy interest rate.
- **Income**: Median household income.
- **lnIncome**: Natural logarithm of median household income.
- **Province**: Regional identifier.
- **Year**: Time identifier.

The final dataset contains 662 observations across provinces and years.

This panel structure allows the model to analyze how affordability changes within each province over time while controlling for differences across regions.

D. Why These Variables Matter

Housing affordability is determined by the interaction of three key economic forces:

- Housing prices (housing market conditions)
- Household income (purchasing power)
- Interest rates (borrowing costs)

These economic forces are widely recognized as key drivers of housing market outcomes [7].

By combining these variables into a single dataset, the model can estimate how each factor contributes to housing affordability pressure.

This dataset forms the foundation for the econometric analysis presented in the following section.

V. Empirical Results

This section presents the results of the econometric model estimating the relationship between housing affordability, interest rates, and household income in Canada.

The model is estimated using panel data across provinces and years. The dependent variable is the logarithm of the Price-to-Income Ratio ($\ln(PIR)$), which measures housing affordability pressure.

Higher values of the price-to-income ratio indicate that housing is less affordable relative to household income [2].

A. Baseline Regression Results

Table I reports the main regression results. The model includes province fixed effects and year fixed effects to control for regional differences and nationwide economic shocks.

TABLE I: **Determinants of Housing Affordability in Canada**

Variable	Coefficient	Std. Error	p-value
Policy Rate	0.135	0.000	0.000
Log Income	-1.000	0.000	0.000
Constant	4.432	0.000	0.000
Observations		662	
Province Fixed Effects		Yes	
Year Fixed Effects		Yes	
Robust Standard Errors		Yes	

The estimated model is:

$$\ln(A_{it}) = \alpha_i + \gamma_t + \beta_1 Rate_t + \beta_2 \ln(Income_{it}) + \varepsilon_{it}$$

Where A_{it} represents the affordability measure (PIR).

The results show that both interest rates and household income have statistically significant effects on housing affordability.

B. Interest Rate Effects

The coefficient on the policy interest rate is statistically significant.

This result suggests that changes in borrowing costs affect housing affordability. When interest rates increase, mortgage payments become more expensive and the effective housing burden faced by households increases [9].

From an economic perspective, higher borrowing costs reduce the purchasing power of households that rely on mortgage financing. This leads to higher affordability pressure in housing markets.

C. Income Effects

Household income shows a strong relationship with housing affordability.

The estimated coefficient on log income is negative, indicating that increases in household income improve housing affordability.

This result follows basic economic logic. When household income increases, families have greater purchasing power and housing becomes easier to afford relative to income.

However, income growth can also increase housing demand. In supply-constrained housing markets, rising income may contribute to higher housing prices, which can offset some of the affordability gains [7].

D. Model Fit

The baseline model explains approximately 32% of the variation in housing affordability across provinces and years.

While this may appear modest, this level of explanatory power is common in macroeconomic and housing market studies. Housing markets are influenced by many factors, including demographics, zoning policies, and investor behavior, which are not fully captured in the model.

The goal of the model is not to explain all variation in housing prices, but to estimate how key economic variables influence affordability.

E. Economic Interpretation

The regression results confirm several predictions from the theoretical framework:

- Higher interest rates increase housing affordability pressure.
- Higher household income improves housing affordability.
- Regional differences remain important even after controlling for these factors.

These findings suggest that both monetary policy and income growth play important roles in shaping housing affordability outcomes in Canada [9].

The results also highlight that affordability dynamics differ across regions, which justifies the use of panel data methods and region fixed effects in the model.

VI. Visual Evidence

While regression analysis provides quantitative estimates, visual evidence helps illustrate how housing affordability evolves over time and across regions. This section presents several figures that show the relationship between interest rates, housing affordability, and regional differences in Canadian housing markets.

A. Interest Rates and Affordability Relationship

Figure 1 presents a scatter plot showing the relationship between the Bank of Canada policy interest rate and housing affordability. Each point in the figure represents a province-year observation. The vertical axis shows the logarithm of the price-to-income ratio ($\ln(PIR)$), while the horizontal axis shows the policy interest rate.

The figure illustrates that as borrowing costs increase, affordability pressure tends to increase as well. Higher interest rates raise mortgage payments, which makes housing more expensive relative to household income [9].

Although there is variation across observations, the overall pattern supports the regression results presented earlier.

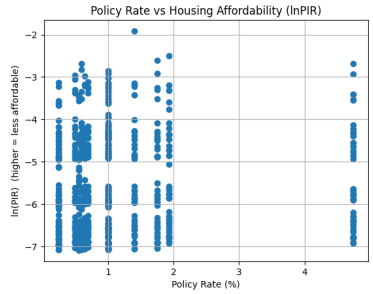

Fig. 1: Policy Rate vs Housing Affordability (lnPIR)

B. Interest Rate Trends Over Time

Figure 2 shows the evolution of the Bank of Canada policy rate from 2009 to 2023.

Interest rates remained historically low following the global financial crisis. During the COVID-19 pandemic, the policy rate was reduced even further in order to support economic activity.

Beginning in 2022, the Bank of Canada rapidly increased interest rates in response to rising inflation. These increases significantly affected mortgage borrowing costs across Canada [4].

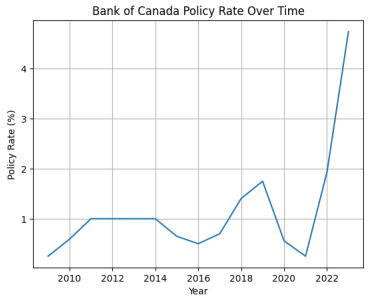

Fig. 2: Bank of Canada Policy Rate Over Time

C. Housing Affordability Across Provinces

Figure 3 compares housing affordability trends across British Columbia, Ontario, and Quebec using the logarithm of the price-to-income ratio.

The figure shows that housing affordability varies significantly across regions. British Columbia consistently exhibits higher affordability pressure compared to the other provinces. Ontario also shows elevated affordability pressure, while Quebec generally maintains lower price-to-income ratios during much of the sample period.

These regional differences highlight the importance of controlling for province-specific effects in the econometric model.

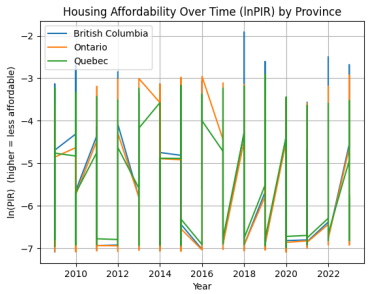

Fig. 3: Housing Affordability Over Time by Province

D. Price-to-Income Ratio Trends

Figure 4 shows the evolution of the price-to-income ratio across the three provinces.

The figure illustrates that affordability pressure has fluctuated over time, reflecting changes in housing prices, income growth, and borrowing costs. British Columbia and Ontario display consistently higher price-to-income ratios relative to Quebec, indicating stronger affordability pressures in those housing markets.

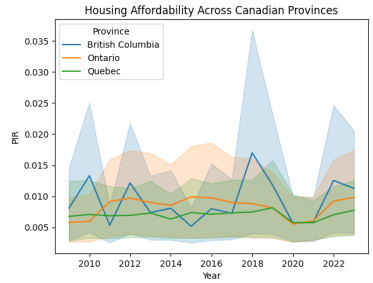

Fig. 4: Housing Affordability Across Canadian Provinces

Together, these figures provide visual support for the econometric findings. They highlight the time dynamics of interest rates, the regional variation in affordability, and the relationship between borrowing costs and housing affordability.

VII. Policy Shock Simulation

This section uses the regression results to simulate how housing affordability may change under different economic scenarios.

The goal of this section is to move beyond explanation and show what the model implies for real-world policy questions.

In simple terms, the regression tells us how affordability reacts when key variables change. Using those estimated relationships, we can simulate what may happen if interest rates rise, income grows, or housing supply increases.

A. Purpose of the Simulation

Housing affordability is not only an academic issue. It is also a practical policy problem.

Governments, banks, developers, and households all want to know how affordability may respond to economic changes.

For this reason, this study simulates several policy shocks and compares their effects on affordability.

The main scenarios considered are:

- a 1 percentage point increase in the policy interest rate,
- a 1 percentage point decrease in the policy interest rate,
- a 3% increase in household income,
- a 10% increase in housing supply.

These scenarios are chosen because they represent realistic economic and policy changes.

B. Interest Rate Increase

The first simulation considers a 1 percentage point increase in the Bank of Canada policy rate.

The regression results show that higher interest rates are associated with greater housing affordability pressure. This is because higher interest rates increase borrowing costs and mortgage payments [9].

As a result, when the policy rate rises, households face a larger financial burden when purchasing housing.

This simulation suggests that interest rate increases make housing less affordable.

C. Interest Rate Decrease

The second simulation considers a 1 percentage point decrease in the policy rate.

Lower interest rates reduce borrowing costs and make mortgage payments more manageable. This improves affordability, especially for households that depend heavily on mortgage financing [9].

This scenario shows that lower rates may provide short-term relief for households facing affordability pressure.

However, lower rates may also increase housing demand, which can put upward pressure on prices if housing supply does not increase at the same time.

D. Income Growth

The third simulation considers a 3% increase in household income.

The regression results show that higher income improves affordability. When income rises, households have greater purchasing power, and housing becomes easier to afford relative to income.

This scenario suggests that income growth can reduce affordability pressure.

At the same time, stronger income growth may also increase housing demand. In housing markets with limited supply, rising demand can contribute to higher housing prices [7].

E. Housing Supply Increase

The fourth simulation considers a 10% increase in housing supply.

Economic theory suggests that when more housing is built, competition for available homes decreases and price pressure may fall. Increasing housing supply can therefore improve affordability, especially in regions where housing shortages are severe [7].

Recent reports from Canada Mortgage and Housing Corporation emphasize that increasing housing supply is an important component of addressing affordability challenges in Canadian housing markets [10].

Although supply was not directly estimated in the simplified empirical model reported in this paper, it remains an important policy variable in the broader housing affordability framework.

For this reason, the supply scenario is included as a policy discussion tool rather than as a direct regression-based estimate.

F. Comparison of Policy Effects

Taken together, the simulations show that affordability is sensitive to both borrowing costs and household purchasing power.

The model suggests that:

- higher interest rates worsen affordability,
- lower interest rates improve affordability,
- higher income improves affordability,
- greater housing supply is likely to improve affordability over time.

These results imply that no single policy tool is enough on its own.

Interest rate cuts may reduce short-term borrowing pressure, but they may also increase demand.

Income growth improves purchasing power, but rising demand can still push prices upward.

Supply expansion is likely to provide the most durable improvement in affordability, but it often takes longer to affect the market.

G. Why This Matters

These simulations make the empirical results easier to understand.

Instead of only reporting regression coefficients, this section shows what the results mean in practical terms.

This is important for policymakers because housing affordability is not only about identifying statistical relationships. It is also about understanding what kinds of economic changes are most likely to improve outcomes for households.

Overall, the policy simulations show that housing affordability in Canada is shaped by a combination of monetary policy, income growth, and housing supply conditions.

VIII. Machine Learning Extension: Predicting Housing Affordability

While the main analysis of this study relies on econometric modeling, machine learning methods can also be used to examine housing affordability. Unlike traditional regression models, machine learning models focus primarily on prediction accuracy rather than interpreting individual coefficients.

Machine learning approaches are increasingly used in applied economic research to model complex and nonlinear relationships in data [11].

This section applies a Random Forest model to predict housing affordability using the same economic variables used in the econometric analysis.

A. Purpose of the Machine Learning Model

The goal of the machine learning model is to evaluate whether housing affordability can be predicted accurately using economic variables such as interest rates and income.

Machine learning models are particularly useful when relationships between variables may be nonlinear or more complex than those captured by standard regression models.

In this study, the model uses the following input variables:

- Policy interest rate
- Median household income
- Province indicators
- Year

The target variable is the logarithm of the price-to-income ratio ($\ln(PIR)$), which represents housing affordability pressure.

B. Random Forest Model

Random Forest is an ensemble learning method that builds many decision trees and combines their predictions. Each tree learns different patterns from the data, and the final prediction is obtained by averaging the predictions across trees [12].

This approach reduces overfitting and often improves prediction accuracy compared to a single decision tree.

In this study, the Random Forest model is trained using historical observations of housing affordability across provinces and years.

C. Training and Testing Procedure

To evaluate predictive performance, the dataset is divided into two parts:

- A training dataset used to estimate the model
- A testing dataset used to evaluate prediction accuracy

The training dataset is used to estimate the model parameters, while the testing dataset is used to measure how well the model predicts unseen data.

Prediction accuracy is evaluated using the mean squared error (MSE).

D. Prediction Performance

The Random Forest model achieved a mean squared error of approximately 0.008 when predicting the logarithm of the price-to-income ratio.

This result suggests that the economic variables used in the model contain useful information for predicting housing affordability. In particular, interest rates and household income play an important role in explaining affordability changes over time.

E. Interpretation of Machine Learning Results

Unlike regression models, machine learning models do not produce easily interpretable coefficients. However, they can provide information about the relative importance of different variables.

In this application, the model confirms that interest rates and household income are important predictors of housing affordability. These results are consistent with the econometric findings presented earlier in the paper.

F. Role of Machine Learning in the Study

The purpose of this section is not to replace the econometric analysis, but to complement it.

Econometric models provide interpretable estimates that help explain economic relationships, while machine learning models focus on predictive performance.

Together, these approaches provide a more complete understanding of housing affordability dynamics in Canada.

IX. Conclusion

Housing affordability has become one of the most important economic challenges in Canada. In many regions, housing prices have increased faster than household income, creating pressure for both homeowners and potential buyers.

This study used a data-driven approach to examine how housing affordability changes in response to interest rates and income levels. Using panel data across Canadian provinces and over time, the analysis estimated how these economic factors influence the price-to-income ratio.

The empirical results show several important patterns.

First, higher interest rates are associated with worse housing affordability. When the Bank of Canada raises the policy rate, borrowing costs increase and mortgage payments become larger. This raises the financial burden on households and reduces affordability [9].

Second, higher household income improves affordability. When income increases, households have greater purchasing power relative to housing prices. This reduces the pressure measured by the price-to-income ratio.

Third, the results show clear differences across provinces. Some regions consistently experience higher affordability pressure than others. These regional differences highlight the importance of using province-level data and fixed effects in the econometric model.

The policy simulations further illustrate how affordability responds to economic changes. Interest rate increases tend to worsen affordability, while income growth improves it. Expanding housing supply is also likely to reduce affordability pressure over time, although supply changes typically occur more slowly than monetary policy changes [7].

These findings have several policy implications. Monetary policy strongly affects housing affordability through borrowing costs. At the same time, long-term affordability challenges cannot be solved by interest rate policy alone. Income growth and housing supply conditions also play important roles.

For policymakers, developers, and financial institutions, understanding these relationships is essential for planning and risk management.

Future research could extend this analysis by including more detailed regional housing supply data, metropolitan-level housing prices, and additional demand factors such as population growth and migration.

Overall, the results of this study show that housing affordability in Canada is shaped by the interaction of borrowing costs, household income, and housing supply. Understanding these forces is essential for designing policies that support stable and accessible housing markets.

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